



Optimized intelligent tracking control for a quadrotor unmanned aerial vehicle with actuator failures

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ABSTRACT

This work investigates the implementation of tracking control for a quadrotor unmanned aerial vehicle exposed to disturbances and actuator failures. Cost reduction in control processes is not usually considered in conventional control methods. To overcome this limitation, this work proposes an optimized intelligent control scheme that utilizes adaptive dynamic programming. Firstly, a critic-only structure is employed to learn the nominal control strategies, including two critic neural network update laws, lifting the persistent excitation condition. Subsequently, two specifically designed observers are developed on the radial basis function neural network (RBFNN) to reconstruct the disturbances and actuator failures and compensate for the nominal strategies. The observers achieve fixed-time convergence by employing the fixed-time update laws of the RBFNNs. Moreover, two variable vectors using the Gaussian error function are introduced to reduce the impact caused by the observation errors. Additionally, the superior performance of the proposed control scheme is validated through numerical simulations.

1. Introduction

Thanks to their superior ability to take off and land vertically, quadrotor unmanned aerial vehicles (UAVs) are widely used in fields such as delivery logistics [1], mapping [2], maritime cruising [3], etc. Trajectory tracking is crucial for quadrotors during task execution in different scenarios. However, it is challenging to develop effective tracking control strategies due to the underactuation and strongly coupled dynamics of quadrotor UAVs. In practical environments, the presence of disturbances (e.g., wind) can lead to poor performance and damage to the quadrotor's actuators, resulting in the failure of tracking tasks. Moreover, the control cost (e.g., electrical energy) consumed in quadrotor flight is very high. Thus, it is vital to develop an efficient control scheme that strikes a balance between control cost and control performance. Various methods have been developed to handle the control problem of nonlinear systems (e.g., the quadrotor UAV), including sliding mode control (SMC), backstepping control, adaptive control, [4][5][6][7] etc. However, none of these schemes achieves a balance between control cost and control performance, thus leading to needless energy waste.

Because of its effectiveness in reducing the control cost, optimal control has been employed to overcome this limitation. However, a crucial challenge to be overcome in optimal control is solving the intractable Hamilton-Jacobi-Bellman (HJB) equation [8]. In this case, the adaptive dynamic programming (ADP) techniques adopt NNs to provide a feasible approach. Therefore, the ADP technique has drawn significant attention from scholars [9][10][11]. As one of the intelligent methods integrated with reinforcement learning technology, the ADP includes two common structures. The traditional structure of the ADP is the actor-critic (AC) structure, using which Vamvoudakis and Lewis [9] dealt with the optimal control problem of nonlinear systems without unknown system dynamics. To solve the tracking control problem, a control algorithm integrating optimization and backstepping control, involving an actor NN and a critic NN devoted to improving and evaluating the control strategy, respectively, was introduced [12]. Wang et al. [13] built a critic-only structure that was superior to AC in computation reduction. Thereafter, the critic-only structure of the ADP technique has been widely applied. Yang et al. [14] used the critic-only structure to develop an ADP-based control strategy that can be employed to solve the attitude reorientation control problem. Nevertheless, the ADP performs poorly in terms of tackling time-varying signals

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[15], and thus fails to meet the requirement of disturbance attenuation when UAVs fly in complex environments. This indicates that the high-quality control of practical UAV systems cannot be realized only through the ADP technique.

To extend the practical applications of the ADP, control frameworks that combine the ADP with other technologies have been proposed to handle the control problem of the system in disturbance or actuator failure conditions [15][16][17][18]. Furthermore, the ADP has been integrated with other techniques, such as the super twisting algorithm [19] and observers [20], in the control design of the quadrotor, thus proving superior to using the ADP technique alone [21]. To handle the tracking control problem of a quadrotor flying in actual environments, many disturbance attenuations [22][23][24] and fault-tolerant control (FTC) schemes [25][26][27] have been proposed. Notably, observer-based control schemes generally yield good performance in handling lumped disturbances mainly induced by actuator failures. However, many of the existing observers require common assumptions (e.g., bounded disturbances and bounded disturbance derivatives) related to disturbances. Developing observers based on NNs is a feasible approach inspired by the approximation of unknown dynamics or states using NNs [28][29].

Due to their strong fitting abilities, radial basis function NNs (RBFNNs) are usually employed to develop observers. An adaptive FTC scheme, which cannot achieve fast convergence characteristics [30], was developed on an RBFNN and L2 gain for a morphing aircraft. To improve the convergence response, Lin et al. [31] proposed an event-based neural control scheme with finite-time convergence characteristics for a six-rotor UAV. Moreover, Liu et al. [32] developed a fixed-time control scheme on RBFNN for multi-robot systems with actuator failures. Although the convergence time can be free of initial conditions, all weights of the RBFNN should be positive. To address this issue, an RBFNN observer-based control scheme was proposed by Zhang et al. [33]. They implemented an adaptive law for RBFNN that achieved fixed-time convergence. However, the upper limit on convergence time did not eliminate the need for acquiring knowledge of the ideal weight information.

Inspired by the above findings, this work addresses the tracking control problem of a quadrotor UAV exposed to disturbances and actuator failures. Intelligent control technologies, optimization ideas, and fault tolerance are integrated into the design process of the scheme for solving the control problem. The main contributions of this work are outlined as follows:

1) An ADP-based control framework is proposed for a quadrotor UAV with lumped disturbances mainly induced by actuator failures. Two critic NN update laws are developed for lifting the common persistent excitation (PE) condition required in [15][21] to learn the approximate optimal nominal control strategies online.

2) Two fixed-time RBFNN observers are developed to reconstruct lumped disturbances. Superior to asymptotic convergence [30] and finite-time convergence [31], the observers achieve fixed-time convergence by employing the proposed fixed-time RBFNN weight update laws. The proposed fixed-time RBFNN weight update law brings benefits as it decouples the observation error settling time from the ideal RBFNN weights, which is more reasonable compared with the existing research [33].

3) Using the Gaussian error function, two variable vectors are proposed to reduce the impact of observation errors on the nominal system. Compared with the use of the hyperbolic tangent function [20], the variable vectors provide higher approximation accuracy. Additionally, the proposed control framework balances control performance and control cost while ensuring that the control system remains ultimately bounded (UUB).

The rest of this paper is organized as follows: The control problem of the quadrotor under actuator failures is formalized in Section 2. Section 3 introduces the ADP-based FTC algorithm, and Section 4 provides the numerical simulation results and analysis. The conclusions are drawn in Section 5.

Notation: The special subscript ζ denotes the scalar or vector of the position subsystem ($\zeta = w$) or attitude subsystem ($\zeta = z$). $\mathbf{I}_n$ denotes a unit matrix with n dimensions. $\cdot$ denotes the lower bound of $\cdot$. $\mathbf{1}_6$ is a column vector expressed by $\mathbf{1}_6 = [1, 1, 1, 1, 1, 1]^T$. In addition, $[\cdot]^\alpha = \|\cdot\|^\alpha \text{sign}(\cdot)$ with $\cdot$ being an arbitrary vector and α a positive constant.

2. Problem formulation

2.1. Dynamic model of a quadrotor UAV

A quadrotor UAV comprises a rigid body and four motors. The body fixed frame $\mathcal{B} = \{x^b, y^b, z^b\}$ and the inertial frame $\mathcal{I} = \{x^e, y^e, z^e\}$ are defined to describe the quadrotor conveniently. $\mathbf{X} = [x, y, z]^T$ denotes the center position of the quadrotor in $\mathcal{I}$. Ω_i is the speed of the i th motor ($i = 1, 2, 3, 4$), which is described in [27]. To illustrate the attitude information in $\mathcal{I}$, the Euler angle vector $\Theta = [\phi, \theta, \psi]^T$ is introduced. The dynamic model of the quadrotor can be established in the following form [34]:

$$m\ddot{\mathbf{X}} + mg\mathbf{z}_I = \mathcal{R}\rho_1\mathbf{u}_t + \mathbf{d}, \quad (1)$$

$$\mathbf{J}(\Theta)\ddot{\Theta} + \mathbf{C}(\Theta, \dot{\Theta})\dot{\Theta} = \rho_a\mathbf{u}_a + \boldsymbol{\tau}_d, \quad (2)$$

where m is the total mass of the quadrotor and g denotes the acceleration of gravity. A unit vector is defined as $\mathbf{z}_I = [0, 0, 1]^T$. The coupling term is denoted as $\mathcal{R} = [\cos\phi\sin\theta\cos\psi + \sin\phi\sin\theta\sin\psi, \cos\phi\sin\theta\sin\psi - \sin\phi\cos\theta, \cos\phi\cos\theta]^T$. $\rho_a = \text{diag}\{\rho_2, \rho_3, \rho_4\}$ with $0 < \rho_i \leq 1$ being the i th control efficiency factor. The definitions of the symmetric matrix $\mathbf{J}(\Theta)$ and Coriolis term $\mathbf{C}(\Theta, \dot{\Theta})$ can be seen in reference [20]. $\mathbf{d} \in \mathbb{R}^3$ and $\boldsymbol{\tau}_d \in \mathbb{R}^3$ are the disturbance forces and torques, respectively, imposed on the position and attitude loops. $\mathbf{u}_t$ and $\mathbf{u}_a = [u_\phi, u_\theta, u_\psi]^T$ are the thrust and torques generated by four rotors, respectively. In addition, the relationship between control inputs (i.e., $\mathbf{u}_t$ and $\mathbf{u}_a$) and the motor speed Ω_i can be seen in [27].

To implement trajectory tracking, the error subsystems should be established subsequently. Define the position error vector $\chi_{w1} = \mathbf{X} - \mathbf{X}_d = [x_e, y_e, z_e]^T$, with $\mathbf{X}_d = [x_d, y_d, z_d]^T$ denoting the desired position. Redefining $\mathbf{v}_w = \mathcal{R}\mathbf{u}_t$ and $\mathbf{v}_z = \mathbf{u}_a$ for convenience, the system (1) can be transformed into the following position error subsystem:

$$\begin{cases} \dot{\chi}_{w1} = \chi_{w2} \\ \dot{\chi}_{w2} = -g\mathbf{z}_I + \frac{1}{m}\mathbf{v}_w + \frac{1}{m}\mathbf{d}_w - \ddot{\mathbf{X}}_d \end{cases}, \quad (3)$$

with lumped disturbance force $\mathbf{d}_w$ being expressed by $\mathbf{d}_w = (\rho_1 - 1)\mathbf{v}_w + \mathbf{d}$.

Generally, $\mathbf{X}$ and ψ are selected as the controlled states to implement the full flight control of the quadrotor, and the desired roll and pitch angles can be obtained by utilizing the determined $\mathbf{X}_d$ and yaw angle ψ_d , which are formulated as follows [21][34]:

$$\begin{aligned} \phi_d &= \arcsin\left(\frac{u_{w1}\sin\psi_d - u_{w2}\cos\psi_d}{\|\mathbf{u}_w\|}\right), \\ \theta_d &= \arctan\left(\frac{u_{w1}\cos\psi_d + u_{w2}\sin\psi_d}{u_{w3}}\right), \end{aligned}$$

with $\mathbf{u}_w = [u_{w1}, u_{w2}, u_{w3}]^T = \frac{1}{m}\mathbf{v}_w$.

The attitude error vector is denoted as $\chi_{z1} = \Theta - \Theta_d = [\phi_e, \theta_e, \psi_e]^T$. Similar to the modeling process for the position error subsystem, the attitude error subsystem can be obtained using

$$\begin{cases} \dot{\chi}_{z1} = \chi_{z2} \\ \dot{\chi}_{z2} = -\mathbf{J}^{-1}\mathbf{C}\dot{\Theta} + \mathbf{J}^{-1}\mathbf{v}_z + \mathbf{J}^{-1}\mathbf{d}_z - \ddot{\Theta}_d \end{cases}, \quad (4)$$

with lumped disturbance torque $\mathbf{d}_z$ being expressed by $\mathbf{d}_z = (\rho_m - \mathbf{I}_3)\mathbf{v}_z + \boldsymbol{\tau}_d$.

2.2. Model transformation

Divide the control laws $\mathbf{v}_\zeta$ into the feed-forward control laws $\mathbf{v}_{\zeta 1}$ and feedback laws $\mathbf{v}_{\zeta 2}$ (e.g., $\mathbf{v}_\zeta = \mathbf{v}_{\zeta 1} + \mathbf{v}_{\zeta 2}$). Two feed-forward control laws are designed as $\mathbf{v}_{z1} = \mathbf{J}\dot{\boldsymbol{\Theta}}_d + \mathbf{C}\boldsymbol{\Theta}_d$ and $\mathbf{v}_{w1} = m\mathbf{g}\mathbf{z}_1 + m\ddot{\mathbf{X}}_d$ to compensate for time-varying known information related to $\mathbf{X}_d$ and $\boldsymbol{\Theta}_d$. Then systems (3) and (4) can be rewritten as

$$\begin{cases} \dot{\mathbf{x}}_{w1} = \mathbf{x}_{w2} \\ \dot{\mathbf{x}}_{w2} = \frac{1}{m}\mathbf{v}_{w2} + \frac{1}{m}\mathbf{d}_w, \end{cases} \quad (5)$$

$$\begin{cases} \dot{\mathbf{x}}_{z1} = \mathbf{x}_{z2} \\ \dot{\mathbf{x}}_{z2} = -\mathbf{J}^{-1}\mathbf{C}\mathbf{x}_{z2} + \mathbf{J}^{-1}\mathbf{v}_{z2} + \mathbf{J}^{-1}\mathbf{d}_z. \end{cases} \quad (6)$$

For simplification, systems (5) and (6) can be rewritten as the following compact form:

$$\dot{\chi}_\zeta = p_\zeta(\chi_\zeta) + q_\zeta(\mathbf{v}_{\zeta 2} + \mathbf{d}_\zeta), \quad (7)$$

with p_ζ and q_ζ being expressed by $p_w(\mathbf{x}_w) = [\mathbf{x}_{w2}^T, \mathbf{0}_{3 \times 1}^T]^T$, $\mathbf{q}_w = [\mathbf{0}_{3 \times 3}, \frac{1}{m}\mathbf{I}_3]^T$, $p_z(\mathbf{x}_z) = [\mathbf{x}_{z2}^T, -(\mathbf{J}^{-1}\mathbf{C}\mathbf{x}_{z2})^T]^T$ and $\mathbf{q}_z = [\mathbf{0}_{3 \times 3}, \mathbf{J}^{-1}]^T$. It is noteworthy that the conditions $p_\zeta(\mathbf{0}) = \mathbf{0}$ are satisfied due to the implementation of the feed-forward control technique.

Therefore, the objective is transformed into finding the feedback control law $\mathbf{v}_{\zeta 2}$ to stabilize the system (7).

Assumption 1. For $\chi_\zeta \in \Phi_\zeta$, there are some positive constants $\check{\delta}_\zeta$, $\check{\sigma}_\zeta$, $\check{q}_\zeta$, $\check{\epsilon}_\zeta$, $\check{\epsilon}_{H\zeta}$ and $\check{\epsilon}_{H\zeta}$ such that $\|\nabla \delta_\zeta(\chi_\zeta)\| \leq \check{\delta}_\zeta$, $\|\sigma_\zeta(\chi_\zeta)\| \leq \check{\sigma}_\zeta$, $\|q_\zeta\| \leq \check{q}_\zeta$, $\|\epsilon_\zeta(\chi_\zeta)\| \leq \check{\epsilon}_\zeta$, $\|\nabla \epsilon_\zeta(\chi_\zeta)\| \leq \check{\epsilon}_{H\zeta}$, and $\|\epsilon_{H\zeta}\| \leq \check{\epsilon}_{H\zeta}$.

3. Optimized intelligent FTC algorithm design

3.1. ADP-based nominal control algorithm

The nominal system of (7) is defined as follows:

$$\dot{\chi}_\zeta = p_\zeta(\chi_\zeta) + q_\zeta \mathbf{v}_{\zeta n}, \quad (8)$$

where $\mathbf{v}_{\zeta n}$ denote the approximate optimized nominal control laws to be developed. By referring to [9][11][15], the cost functions of the nominal system (8) can be defined in the following form:

$$J_\zeta(\chi_\zeta) = \int_0^\infty \chi_\zeta^T \mathbf{Q}_\zeta \chi_\zeta + \mathbf{v}_{\zeta n}^T \mathbf{R}_\zeta \mathbf{v}_{\zeta n} d\theta. \quad (9)$$

Assuming that the optimal control strategies $\mathbf{v}_{\zeta n}^*$ exist, the corresponding $J_\zeta^*(\chi_\zeta)$ can be yielded as:

$$J_\zeta^*(\chi_\zeta) = \int_0^\infty \chi_\zeta^T \mathbf{Q}_\zeta \chi_\zeta + \mathbf{v}_{\zeta n}^{*T} \mathbf{R}_\zeta \mathbf{v}_{\zeta n}^* d\theta. \quad (10)$$

Taking the time derivative of (10), one has:

$$H_\zeta(\chi_\zeta, \mathbf{v}_{\zeta n}^*, \nabla J_\zeta^*(\chi_\zeta)) = \nabla^T J_\zeta^*(\chi_\zeta) (p_\zeta(\chi_\zeta) + q_\zeta \mathbf{v}_{\zeta n}^*) + \chi_\zeta^T \mathbf{Q}_\zeta \chi_\zeta + \mathbf{v}_{\zeta n}^{*T} \mathbf{R}_\zeta \mathbf{v}_{\zeta n}^* = 0. \quad (11)$$

Then, the optimal control policies $\mathbf{v}_{\zeta n}^*$ can be calculated using:

$$\mathbf{v}_{\zeta n}^* = -\frac{1}{2} \mathbf{R}_\zeta^{-1} q_\zeta^T \nabla J_\zeta^*(\chi_\zeta). \quad (12)$$

Substituting $\mathbf{v}_{\zeta n}^*$ into (11), the HJB equations can be yielded as:

$$\chi_\zeta^T \mathbf{Q}_\zeta \chi_\zeta + \nabla^T J_\zeta^* p_\zeta(\chi_\zeta) - \frac{1}{4} \nabla^T J_\zeta^*(\chi_\zeta) q_\zeta \mathbf{R}_\zeta^{-1} q_\zeta^T \nabla J_\zeta^*(\chi_\zeta) = 0. \quad (13)$$

Obtaining analytically exact solutions for the HJB equations is widely acknowledged as very challenging. It is worth noting that the

NN can provide a superior scheme for this problem. If NNs contain abundant sets of basis functions, the optimal cost function can be reconstructed as follows:

$$J_\zeta^*(\chi_\zeta) = \omega_\zeta^T \delta_\zeta(\chi_\zeta) + \epsilon_\zeta(\chi_\zeta), \quad (14)$$

where $\chi_\zeta \in \mathbb{C}_\zeta$ are compact sets. Thus, the optimal control policy $\mathbf{v}_{\zeta n}^*$ can be converted below:

$$\mathbf{v}_{\zeta n}^* = -\frac{1}{2} \mathbf{R}_\zeta^{-1} q_\zeta^T (\nabla^T \delta_\zeta(\chi_\zeta) \omega_\zeta + \nabla \epsilon_\zeta(\chi_\zeta)). \quad (15)$$

Owing to the unknown optimal weight vectors, the approximation results of (14) can be obtained by:

$$J_\zeta(\chi_\zeta) = \hat{\omega}_\zeta^T \delta_\zeta(\chi_\zeta). \quad (16)$$

Similarly, the optimal control policy (15) can be approximated as:

$$\mathbf{v}_{\zeta n} = -\frac{1}{2} \mathbf{R}_\zeta^{-1} q_\zeta^T \nabla^T \delta_\zeta(\chi_\zeta) \hat{\omega}_\zeta. \quad (17)$$

Subsequently, the approximated nonlinear functions can be derived below:

$$H_\zeta(\chi_\zeta, \mathbf{v}_{\zeta n}, \hat{\omega}_\zeta) = \hat{\omega}_\zeta^T \gamma_\zeta + \chi_\zeta^T \mathbf{Q}_\zeta \chi_\zeta + \mathbf{v}_{\zeta n}^T \mathbf{R}_\zeta \mathbf{v}_{\zeta n} = e_{\zeta 1} \quad (18)$$

with $\gamma_\zeta = \nabla \delta_\zeta(p_\zeta(\chi_\zeta) + q_\zeta \mathbf{v}_{\zeta n})$. According to [9], $e_{\zeta 1}$ can be rewritten as:

$$e_{\zeta 1} = \tilde{\omega}_\zeta^T \gamma_\zeta + \epsilon_{\zeta H}, \quad (19)$$

where $\tilde{\omega}_\zeta = \hat{\omega}_\zeta - \omega_\zeta$, and the definitions of $\epsilon_{\zeta H}$ are consistent with [9]. It is noteworthy that the relationship between $e_{\zeta 1}$ and the weight approximation errors $\tilde{\omega}_\zeta$ can be inferred from (19). Thus, $e_{\zeta 1}$ is generally used when developing critic NN weight update laws.

Theorem 1. For nominal system (8) with cost function (9), if control strategies $\mathbf{v}_{\zeta n}$ are given as (17), and the critic NN update laws are designed as follows:

$$\dot{\hat{\omega}}_\zeta = -\kappa_{\zeta 1} \frac{e_{\zeta 1} \gamma_\zeta}{(1 + \gamma_\zeta^T \gamma_\zeta)^2} - \kappa_{\zeta 2} \varphi_\zeta \hat{\omega}_\zeta, \quad (20)$$

with $\kappa_{\zeta 1}, \kappa_{\zeta 2} \in \mathbb{R}^+$ and $\varphi_\zeta = (\|\nabla \delta_\zeta(\chi_\zeta)\| + \mu_\zeta l_\zeta)$. μ_ζ denote small positive constants. Besides, l_ζ are switching quantities (e.g., if $\|\nabla \delta_\zeta(\chi_\zeta)\| = 0$, then $l_\zeta = 1$; otherwise, $l_\zeta = 0$). Then, the UUB stability of $\tilde{\omega}_\zeta$ and χ_ζ in (8) is guaranteed.

Proof. See Subsection 6.1 for details.

One can conclude that the proposed control policies (17) can guarantee the UUB stability of the nominal systems (8). The nominal control laws are baseline control laws that can stabilize nominal systems. However, to tackle the optimized tracking problem of practical systems modeled in (7), it is necessary to carry out the FTC scheme in the following design process.

3.2. Fixed-time RBFNN observer for the lumped disturbances

The ADP technique lacks the ability to handle time-varying signals, resulting in poor tracking performance when the quadrotor faces lumped disturbances. In order to overcome this defect, two disturbance observers are developed to compensate for the lumped disturbances. Owing to the strong approximation characteristics of the RBFNN, lumped disturbance terms $\mathbf{d}_\zeta$ can be reconstructed using the following equations:

$$\mathbf{d}_\zeta = \mathbf{W}_\zeta^T \sigma_\zeta(\chi_\zeta) + \epsilon_\zeta(\chi_\zeta), \quad (21)$$

where $\mathbf{W}_\zeta \in \mathbb{R}^{f \times t}$ denote the ideal weight matrices of two RBFNNs. The radial basis functions are denoted as

$$\sigma_\zeta(\chi_\zeta) = [\sigma_{\zeta 1}(\chi_\zeta), \sigma_{\zeta 2}(\chi_\zeta), \dots, \sigma_{\zeta f}(\chi_\zeta)]^T$$

(e.g., $\sigma_{\zeta f}(\chi_\zeta) = \exp(-\frac{\|\chi_\zeta - \mathbf{Y}_\zeta\|^2}{\eta_{\zeta f}^2})$, $f = 1, \dots, 6$). $\eta_{\zeta f}$ refer to the width

of the Gaussian function, $\mathbf{Y}_\zeta = [\mathbf{Y}_{\zeta 1}, \mathbf{Y}_{\zeta 2}, \dots, \mathbf{Y}_{\zeta 6}]^T$ represent the center vectors of the receptive field, and $\varepsilon_\zeta(\chi_\zeta) \in \mathbb{R}^6$ denote the approximation errors.

Due to the unknown ideal weights $\mathbf{W}_\zeta$, the fault information can be approximated as follows:

$$\hat{\mathbf{d}}_\zeta = \hat{\mathbf{W}}_\zeta^T \sigma_\zeta(\chi_\zeta), \quad (22)$$

where the estimation results for $\mathbf{W}_\zeta$ are represented by $\hat{\mathbf{W}}_\zeta$. The weight approximation errors, $\tilde{\mathbf{W}}_\zeta$, and observation errors, $\tilde{\mathbf{d}}_\zeta$, can be expressed as $\tilde{\mathbf{W}}_\zeta = \hat{\mathbf{W}}_\zeta - \mathbf{W}_\zeta$ and $\tilde{\mathbf{d}}_\zeta = \mathbf{d}_\zeta - \hat{\mathbf{d}}_\zeta$, respectively. Define the error vector $\mathbf{e}_\zeta = \chi_\zeta - \hat{\chi}_\zeta$ with two auxiliary systems being designed as:

$$\dot{\hat{\chi}}_\zeta = \lambda_{\zeta 1} \mathbf{e}_\zeta + \lambda_{\zeta 2} [\mathbf{e}_\zeta]^{\frac{1}{2}} + \lambda_{\zeta 3} [\mathbf{e}_\zeta]^3 + p_\zeta(\chi_\zeta) + \mathbf{q}_\zeta \mathbf{v}_{\zeta 2} + \mathbf{q}_\zeta \tilde{\mathbf{d}}_\zeta. \quad (23)$$

Taking the derivative of the aforedefined $\mathbf{e}_\zeta$ and recalling (7), one has

$$\dot{\mathbf{e}}_\zeta = -\lambda_{\zeta 1} \mathbf{e}_\zeta - \lambda_{\zeta 2} [\mathbf{e}_\zeta]^{\frac{1}{2}} - \lambda_{\zeta 3} [\mathbf{e}_\zeta]^3 - \mathbf{q}_\zeta (\tilde{\mathbf{W}}_\zeta^T \sigma_\zeta(\chi_\zeta) - \varepsilon_\zeta(\chi_\zeta)). \quad (24)$$

Theorem 2. Consider the RBFNN observers designed in (22), if the weight adaptive laws of RBFNNs are designed below:

$$\dot{\tilde{\mathbf{W}}}_\zeta = -a_\zeta (-\sigma_\zeta(\chi_\zeta) \mathbf{e}_\zeta^T \mathbf{q}_\zeta + b_\zeta \tilde{\mathbf{W}}_\zeta + c_\zeta \tilde{\mathbf{W}}_\zeta \tilde{\mathbf{W}}_\zeta^T \tilde{\mathbf{W}}_\zeta), \quad (25)$$

with $a_\zeta \in \mathbb{R}^+$. b_ζ and c_ζ are positive constants. Then, the observation errors can achieve practical fixed-time stability.

Proof. See Subsection 6.2 for details.

3.3. ADP-based optimized intelligent FTC algorithm

Integrating the ADP-based nominal controllers and RBFNN observers, the optimized intelligent FTC scheme will be developed in this subsection.

Theorem 3. For practical systems in (7), if two control laws are designed as follows:

$$\mathbf{v}_{\zeta 2} = \frac{1}{2} \mathbf{v}_{\zeta n} - \hat{\mathbf{d}}_\zeta - \mathbf{t}_\zeta, \quad (26)$$

where variable vectors $\mathbf{t}_\zeta$ are expressed as $\mathbf{t}_\zeta = \varpi_\zeta \mathbf{q}_\zeta^+ \text{erf}(\nabla J_\zeta / \rho_\zeta)$, with ϖ_ζ denoting the positive constants. Then, the tracking errors χ_ζ in practical systems (7) and weight errors $\tilde{\mathbf{w}}_\zeta$ are UUB stable.

Proof. See Subsection 6.3 for details.

Remark 1. Since the approximation errors the critic NNs cannot converge at the origin, it is almost impossible for $\|\nabla \delta_\zeta(\chi_\zeta)\| = 0$ in Theorem 1 to hold. However, once the above conditions hold, the bounds of approximation errors cannot be obtained. To solve this problem, switching quantities are designed. To be specific, if the conditions hold, the bounds of approximation errors can be obtained. On the other hand (e.g., $\|\nabla \delta_\zeta(\chi_\zeta)\| \neq 0$), the approximation errors can converge simultaneously with the convergence of the system states.

Remark 2. The well-designed continuous terms $\text{erf}(\nabla J_\zeta / \rho_\zeta)$ are employed to attenuate the system impact because of the observation errors. Superior to the continuous robust term constructed by $\tanh(\cdot)$, $\text{erf}(\cdot)$

performs better when it is used for smooth approximations of $|\cdot|$ with respect to accuracy [35]. Moreover, using $\text{erf}(\cdot)$ to replace a discontinuous robust term (e.g., $\text{sign}(\cdot)$) can effectively improve the chattering phenomenon.

4. Simulation results and analysis

Numerical simulation results will be presented in this section to validate the effectiveness of the proposed control scheme (called ADPNN).

4.1. Simulation initial conditions setting

Simulations are performed on a quadrotor UAV. The physical parameters of the quadrotor, the initial conditions, and the numerical values of external disturbances were defined the same as [20]. To simulate actuator failures, mutation-induced control efficiency factors are set, which are described by

$$\rho_1 = \begin{cases} 1, & t < 20 \\ 0.9, & \text{else} \end{cases}, \quad \rho_2 = \begin{cases} 1, & t < 15 \\ 0.9, & \text{else} \end{cases},$$

$$\rho_3 = \begin{cases} 1, & t < 25 \\ 0.9, & \text{else} \end{cases} \quad \text{and} \quad \rho_4 = \begin{cases} 1, & t < 35 \\ 0.9, & \text{else} \end{cases}.$$

The critic NNs are employed to learn the nominal control strategy, and RBFNNs are to reconstruct lumped simulations. The basis functions of the critic NNs are referred to [20]. The initial weights of critic NN are $\hat{\mathbf{w}}_w(0) = [25, 25, 30, 25, 25, 30]^T$ and $\hat{\mathbf{w}}_z(0) = \mathbf{1}_6$. In addition, the main parameters are $\mathbf{R}_w = 2.5 \mathbf{I}_3$, $\mathbf{R}_z = 19 \mathbf{I}_3$, $\mathbf{Q}_w = 5 \mathbf{I}_6$, $\mathbf{Q}_z = 20 \mathbf{I}_6$, $\lambda_{w1} = \lambda_{z1} = 50$, $\lambda_{w2} = 0.001$, $\lambda_{z2} = 0.2$, $\lambda_{w3} = 0.01$, $\lambda_{z3} = 0.3$, $\kappa_{w1} = 0.5$, $\kappa_{z1} = 0.05$, $a_w = 600$, $a_z = 5$, $k_w = 0.5$, and $k_z = 2$.

4.2. Numerical simulations implementation driven by different schemes

To highlight the superiority and effectiveness of the proposed ADPNN scheme, two comparison schemes are implemented. One combines the ADP-based control strategy proposed in [36] with fixed-time NN observers proposed in [33] (called ADPFT), and the other is presented in the Appendix of [37] (called PIDSMC). For a fairer comparison, the convergence time of the tracking errors belonging to different control schemes is set to an almost equal level by adjusting the parameters. Notably, in the comparison simulation figures, the red, purple, and green curves represent the proposed ADPNN, PIDSMC, and ADPFT schemes, respectively. Figs. 1 and 2 illustrate the tracking errors of the quadrotor. It can be seen that the tracking errors can converge into a small region when $t = 5$ s, which indicates that the proposed fixed-time RBFNN observers possess fine performance in estimating lumped disturbances. The nominal control strategies can guarantee the stability of the system when $t > 5$ s. The 3D tracking results shown in Fig. 3 are employed to describe the tracking effect more vividly. From Figs. 1 and 2, one can see that the numerical values of the red curves are the smallest, which indicates that the proposed ADPNN possesses the highest accuracy compared with PIDSMC and ADPFT.

Figs. 4 and 5 show the time responses of the observation errors. The NN observers proposed in both ADPNN and ADPFT schemes can estimate lumped disturbances effectively. But the observers in ADPNN possess higher accuracy, with $\tilde{d}_{w1} - \tilde{d}_{w3}$ being less than 4×10^{-3} N and $\tilde{d}_{z1} - \tilde{d}_{z3}$ being less than 4×10^{-4} N · m. It is evident that the numerical values of the ADPFT scheme are greater than the proposed ADPNN scheme. Therefore, the RBFNN observers proposed in ADPNN can perform better in observation accuracy compared with the observer in ADPFT. The approximation results of the critic NNs are depicted in Fig. 6. It can be found that all the curves of critic NN weights tend to remain stable with almost no fluctuation after 5 s. This is because actuator failures have been compensated by the RBFNN observers. As shown in Fig. 7, the red curves fluctuate faster than both the purple and green

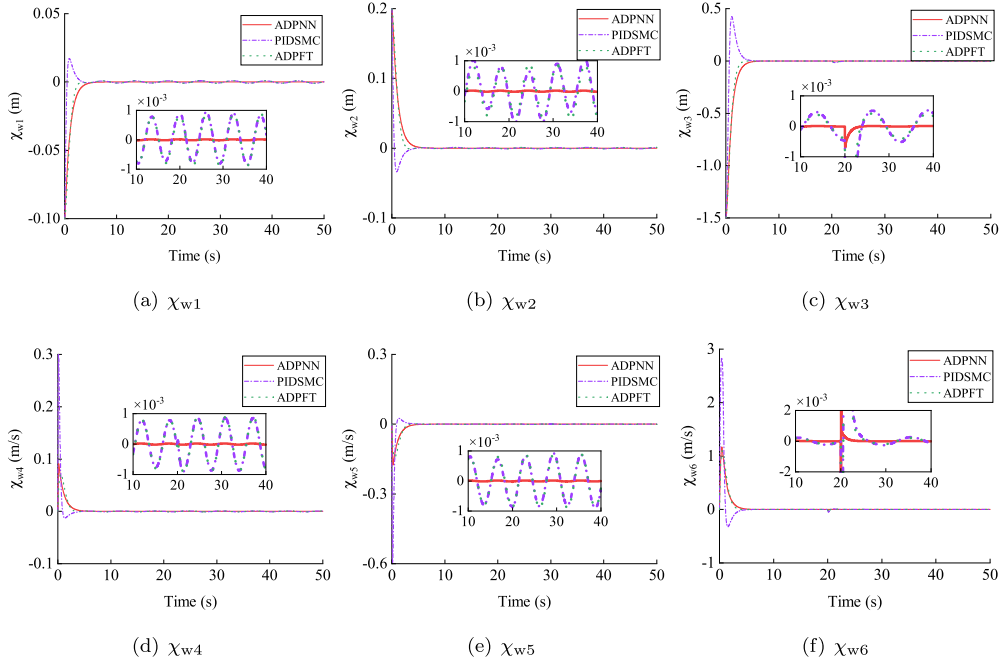


Fig. 1. Tracking error results of the position loop driven by different schemes. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

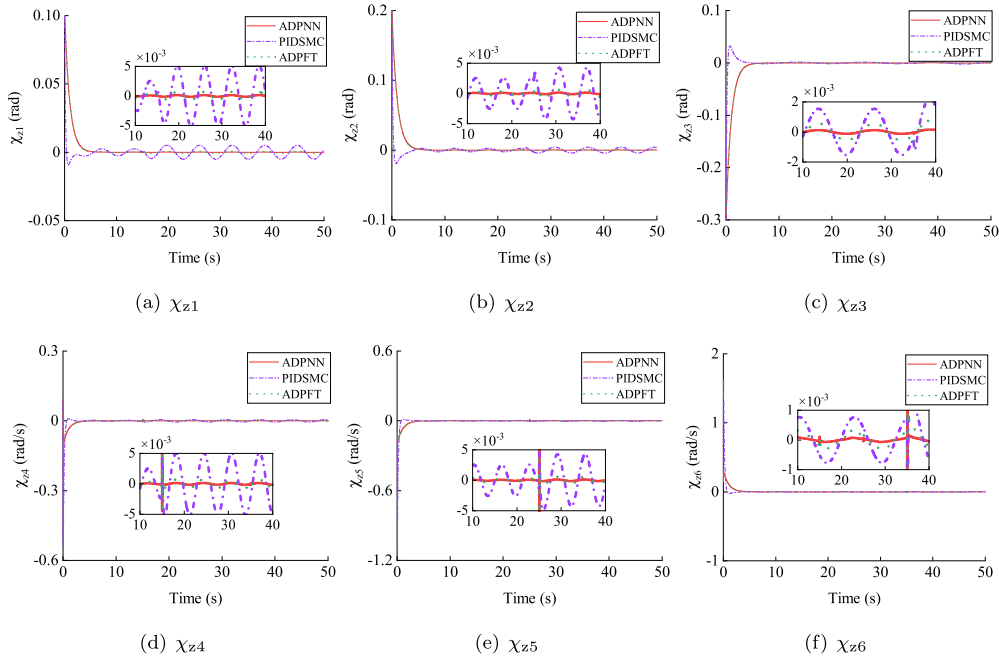


Fig. 2. Tracking error results of the attitude loop driven by different schemes.

curves when $t = 15, 20, 25, 35$ s. This is because the proposed ADPNN can respond quickly to actuator failures under the fixed-time RBFNN observers. Moreover, the values of the purple curves in the initial period are much larger than the red and green curves, demonstrating that traditional control frameworks (i.e., such as PIDSMC) may result in higher control cost.

While ADPNN and ADPFT based on optimal control methods can make a trade-off between control cost and control performance, the comparison in cost reduction between ADPNN and ADPFT is not implemented herein. To highlight the scheme's compelling performances in reducing control cost, let us define two novel cost functions $J_{\zeta}(\chi_{\zeta}) =$

$\int_0^{\infty} (\chi_{\zeta}^T Q_{\zeta} \chi_{\zeta} + v_{\zeta}^T R_{\zeta} v_{\zeta}) d\theta$. Notably, v_{ζ} are the total control laws belonging to the two subsystems, rather than the nominal control strategies. The histograms in Fig. 8 illustrate the comparison results of control cost driven by ADPNN and PIDSMC. One can see that the total cost values under the proposed ADPNN are smaller than those under the PIDSMC used for comparison quantitatively. Moreover, the cost values of the position loop are much larger than the attitude loop; this is because the control inputs shown in Fig. 7 of the position loop are more significant. To be specific, the cost values J_{ζ} under the proposed ADPNN are smaller in the interval 0–15 s, while they are almost equal in the other two intervals (e.g., 15–30 s and 30–50 s) under different

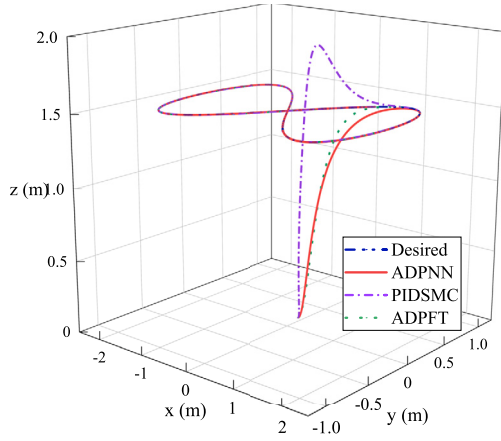


Fig. 3. 3D tracking results driven by different schemes.

control schemes, as shown in Fig. 8. Because the control inputs of the initial stage under the proposed ADPNN are at a higher level, the system error states can converge into a small neighborhood of the origin. From the above discussions and analysis, we can conclude that the proposed ADPNN yields superior performance in striking a balance between control cost and control performance when the quadrotor is exposed to actuator failures.

5. Conclusion

This work developed an optimized tracking control scheme for a quadrotor UAV with actuator failures, which ensures UUB stability of the control system. The critic-only structure was utilized to learn the nominal control strategies through the proposed critic NN-based update laws without the PE condition. On this basis, two RBFNN observers were designed to reconstruct lumped disturbances while guaranteeing fixed-time convergence characteristics. The observers feature upper bounds on the RBFNN weight errors that no longer rely on their ideal weights. Two specifically designed continuous variable vectors with the introduction of the Gaussian error function were employed to enhance the system's control performance. A comparison of the simulation results demonstrated the effectiveness of the proposed scheme in reducing control cost. In the future, the authors will focus on improving their research on fixed-time ADP and its applications to game-based control frameworks.

6. Appendix

6.1. Proof of Theorem 1

Choose Lyapunov functions as follows:

$$V_{\zeta 1} = J_{\zeta}^*(\chi_{\zeta}) + \frac{1}{2} \tilde{\omega}_{\zeta}^T \tilde{\omega}_{\zeta}. \quad (27)$$

Notably, $G_{\zeta} = \nabla \delta_{\zeta} q_{\zeta} R_{\zeta}^{-1} q_{\zeta}^T \nabla^T \delta_{\zeta}$ and $A_{\zeta} = q_{\zeta} R_{\zeta}^{-1} q_{\zeta}^T$, with G_{ζ} and A_{ζ} being symmetric non-negative matrices. The time derivative of (27) is yielded as:

$$\begin{aligned} \dot{V}_{\zeta 1} &= -\chi_{\zeta}^T Q_{\zeta} \chi_{\zeta} - \nu_{\zeta n}^* R_{\zeta} \nu_{\zeta n} \\ &\quad + (\omega^T \nabla \delta_{\zeta} + \nabla^T \epsilon_{\zeta}) q_{\zeta} \left(\frac{1}{2} R_{\zeta}^{-1} q_{\zeta}^T \nabla^T \delta_{\zeta} \omega_{\zeta} \right. \\ &\quad \left. - \frac{1}{2} R_{\zeta}^{-1} q_{\zeta}^T \nabla^T \delta_{\zeta} \tilde{\omega}_{\zeta} + \frac{1}{2} R_{\zeta}^{-1} q_{\zeta}^T \nabla \epsilon_{\zeta} \right) + \tilde{\omega}_{\zeta}^T \dot{\tilde{\omega}}_{\zeta} \\ &\leq -\chi_{\zeta}^T Q_{\zeta} \chi_{\zeta} + \frac{1}{2} \tilde{\omega}_{\zeta}^T G_{\zeta} \tilde{\omega}_{\zeta} \\ &\quad - \tilde{\omega}_{\zeta}^T \left(\frac{\kappa_{\zeta 1} \gamma_{\zeta} \gamma_{\zeta}^T}{2(1 + \gamma_{\zeta}^T \gamma_{\zeta})} + \frac{\kappa_{\zeta 2}}{2} \varphi_{\zeta} I_6 \right) \tilde{\omega}_{\zeta} + \epsilon_{\zeta d}, \end{aligned} \quad (28)$$

where $\epsilon_{\zeta d} = \frac{\kappa_{\zeta 1} \tilde{\epsilon}_{\zeta H}^2}{2(1 + \gamma_{\zeta}^T \gamma_{\zeta})} + \frac{\kappa_{\zeta 2} \tilde{\delta}_{\zeta}}{2} \|\omega_{\zeta}\|^2 + \frac{\tilde{\epsilon}_{\zeta}^2}{2} \lambda_{\max}(A_{\zeta})$. Define $C_{\zeta} = \kappa_{\zeta 2} \varphi_{\zeta} I_6 + \frac{\kappa_{\zeta 1} \gamma_{\zeta} \gamma_{\zeta}^T}{(1 + \gamma_{\zeta}^T \gamma_{\zeta})} - G_{\zeta}$, then

$$\dot{V}_{\zeta 1} \leq -\chi_{\zeta}^T Q_{\zeta} \chi_{\zeta} - \frac{1}{2} \tilde{\omega}_{\zeta}^T C_{\zeta} \tilde{\omega}_{\zeta} + \epsilon_{\zeta d}. \quad (29)$$

By referring to Assumption 1, it can be concluded that $\|\tilde{G}_{\zeta}\|$ is bounded, such that $\|\tilde{G}_{\zeta}\| \leq \tilde{G}_{\zeta}$. Also, if $\lambda_{\min}(C_{\zeta}) > 0$ can be satisfied by selecting appropriate parameters $\kappa_{\zeta 1}$ and $\kappa_{\zeta 2}$, then χ_{ζ} and $\tilde{\omega}_{\zeta}$ are guaranteed to achieve UUB. Therefore, the proposed novel weight update laws demonstrably relax the common PE condition.

6.2. Proof of Theorem 2

Consider Lyapunov functions as follows:

$$V_{\zeta 2} = \mathcal{L}_{\zeta 1} + \mathcal{L}_{\zeta 2} = \frac{1}{2} e_{\zeta}^T e_{\zeta} + \frac{1}{2} \sum_{f=1}^6 \frac{1}{a_{\zeta}} \left(\sum_{i=1}^3 \tilde{W}_{\zeta f i}^T \tilde{W}_{\zeta f i} \right). \quad (30)$$

If positive constants $\lambda_{\zeta 1}$ satisfy $\lambda_{\zeta 1} > 0.5$, then the time derivative of $\mathcal{L}_{\zeta}$ can be obtained in the following form:

$$\dot{\mathcal{L}}_{\zeta 1} \leq -\lambda_{\zeta 2} \|e_{\zeta}\|^{\frac{3}{2}} - \lambda_{\zeta 3} \|e_{\zeta}\|^4 - e_{\zeta}^T q_{\zeta} \tilde{W}_{\zeta}^T \sigma_{\zeta}(\chi_{\zeta}) + \frac{\tilde{q}_{\zeta}^2}{2} \|\epsilon_{\zeta}(\chi_{\zeta})\|^2. \quad (31)$$

Taking the time derivative of $\mathcal{L}_{\zeta 2}$ and according to Lemma 2 in [38], the following inequalities hold:

$$\begin{aligned} \dot{\mathcal{L}}_{\zeta 2} &\leq \sum_{f=1}^6 \left(\sum_{i=1}^3 \left(\tilde{W}_{\zeta f i}^T \sigma_{\zeta f}(\chi_{\zeta}) e_{\zeta f}^T q_{\zeta f i} - \frac{b_{\zeta}}{2} \|\tilde{W}_{\zeta f i}\|^2 + \frac{b_{\zeta}}{2} \|W_{\zeta f i}\|^2 \right. \right. \\ &\quad \left. \left. + \frac{3c_{\zeta} s^{\frac{4}{3}}}{4} \|\tilde{W}_{\zeta f i}\|^4 + \frac{c_{\zeta}}{4s^4} \|\tilde{W}_{\zeta f i}\|^4 + \frac{9c_{\zeta} s^{\frac{4}{3}}}{4} \|\tilde{W}_{\zeta f i}\|^4 \right. \right. \\ &\quad \left. \left. - c_{\zeta} \|\tilde{W}_{\zeta f i}\|^4 + \frac{3c_{\zeta} \tilde{\delta}_{\zeta}}{4s^4} \|\tilde{W}_{\zeta f i}\|^4 \right) \right) \\ &\leq \sum_{f=1}^6 \left(\sum_{i=1}^3 \left(\tilde{W}_{\zeta f i}^T \sigma_{\zeta f}(\chi_{\zeta}) e_{\zeta f}^T q_{\zeta f i} - \frac{b_{\zeta}}{8} \|\tilde{W}_{\zeta f i}\|^2 - \frac{b_{\zeta}}{2} \|\tilde{W}_{\zeta f i}\|^{\frac{3}{2}} \right. \right. \\ &\quad \left. \left. - \left(c_{\zeta} - \frac{c_{\zeta}}{4s^4} - \frac{9c_{\zeta} s^{\frac{4}{3}}}{4} \right) \|\tilde{W}_{\zeta f i}\|^4 + \left(\frac{3c_{\zeta} s^{\frac{4}{3}}}{4} + \frac{3c_{\zeta}}{4s^4} \right) \|W_{\zeta f i}\|^4 \right. \right. \\ &\quad \left. \left. + \frac{b_{\zeta}}{2} \|W_{\zeta f i}\|^2 \right) + \frac{b_{\zeta}}{8} \right). \end{aligned} \quad (32)$$

Referring to Lemmas 3 and 4 in [39], and substituting (31) and (32) back into the time derivatives of (30), the following inequalities can be obtained:

$$\begin{aligned} \dot{V}_{\zeta 2} &\leq -\lambda_{\zeta 2} \|e_{\zeta}\|^{\frac{3}{2}} - \lambda_{\zeta 3} \|e_{\zeta}\|^4 \\ &\quad + \sum_{f=1}^6 \left(\sum_{i=1}^3 \left(-\frac{b_{\zeta}}{8} \|\tilde{W}_{\zeta f i}\|^2 - \frac{b_{\zeta}}{2} \|\tilde{W}_{\zeta f i}\|^{\frac{3}{2}} \right. \right. \\ &\quad \left. \left. - \left(c_{\zeta} - \frac{c_{\zeta}}{4s^4} - \frac{9c_{\zeta} s^{\frac{4}{3}}}{4} \right) \|\tilde{W}_{\zeta f i}\|^4 \right) \right) + \vartheta_{\zeta} \\ &\leq -\tilde{\alpha}_{\zeta} V_{\zeta}^{\frac{3}{4}} - \tilde{\beta}_{\zeta} V_{\zeta}^2 + \vartheta_{\zeta}, \end{aligned} \quad (33)$$

where the expressions of $\tilde{\alpha}_{\zeta}$, $\tilde{\beta}_{\zeta}$, and ϑ_{ζ} are

$$\tilde{\alpha}_{\zeta} = \min \left\{ 2^{\frac{3}{4}} \lambda_{\zeta 2}, 2^{-\frac{1}{4}} a_{\zeta}^{\frac{3}{2}} b_{\zeta} \right\},$$

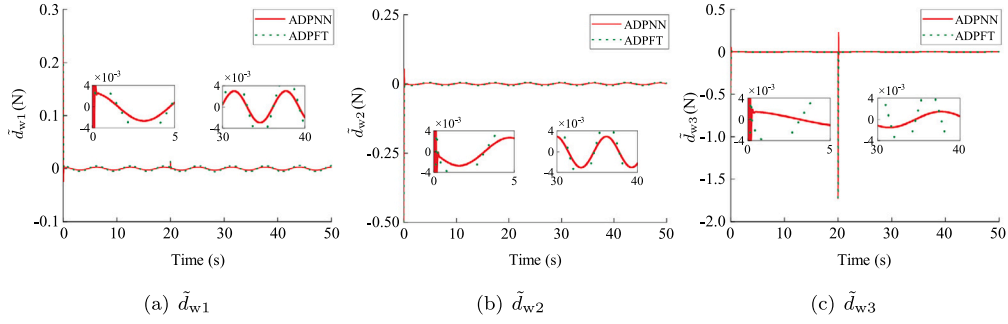


Fig. 4. Observation error results of the position loop driven by different RBFNN observers.

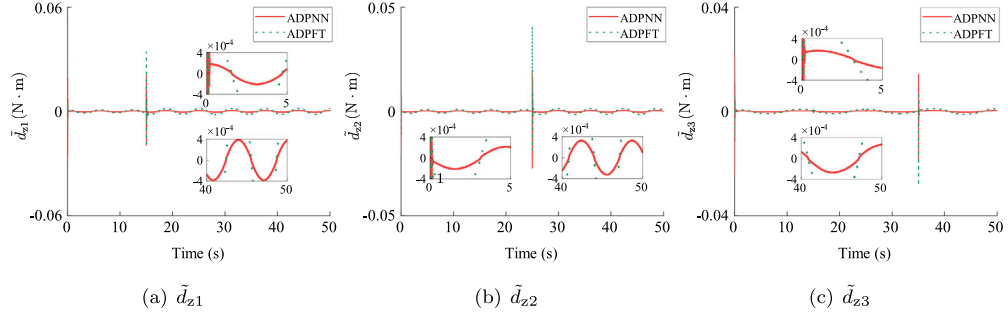


Fig. 5. Observation error results of the attitude loop driven by different RBFNN observers.

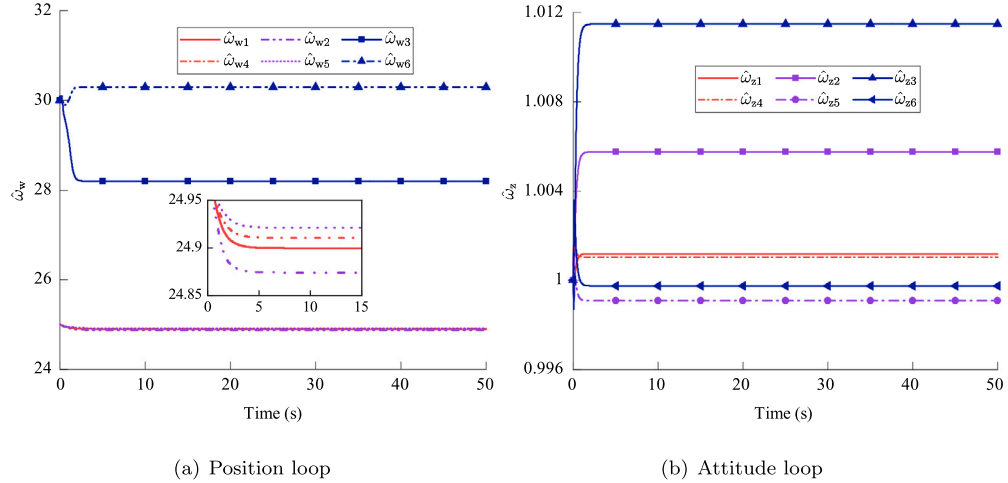


Fig. 6. Weight results of the critic NNs driven by ADPNN.

$$\check{\beta}_\zeta = \min \left\{ 4\lambda_{\zeta 3}, \frac{a_\zeta^2}{9} \left(c_\zeta - \frac{c_\zeta}{4s_\zeta^4} - \frac{9c_\zeta s_\zeta^{\frac{4}{3}}}{4} \right) \right\}$$

and

$$\begin{aligned} \vartheta_\zeta = & \sum_{f=1}^6 \left(\sum_{i=1}^3 \left(\left(\frac{3c_\zeta s_\zeta^{\frac{4}{3}}}{4} + \frac{3c_\zeta}{4s_\zeta^4} \right) \|W_{\zeta f i}\|^4 + \frac{b_\zeta}{2} \|W_{\zeta f i}\|^2 \right) + \frac{b_\zeta}{8} \right) \\ & + \frac{\check{q}_\zeta}{2} \|\varepsilon_\zeta^T(\chi_\zeta)\|^2. \end{aligned}$$

According to Lemma 2 in [4], one can obtain that

$$\left\{ (e_\zeta, \tilde{W}_{\zeta f i}) \mid V_\zeta(e_\zeta, \tilde{W}_{\zeta f i}) \leq \min \left\{ \left(\frac{g_\zeta}{\check{\alpha}_\zeta} \right)^{\frac{4}{3}}, \left(\frac{g_\zeta}{\check{\beta}_\zeta} \right)^{\frac{1}{2}} \right\} \right\}$$

with $g_\zeta = \frac{\vartheta_\zeta}{1-\theta_{\zeta 0}}$ in fixed time $t_{\zeta u} \leq \frac{4}{\check{\alpha}_\zeta \theta_{\zeta 0}} + \frac{1}{\check{\beta}_\zeta \theta_{\zeta 0}}$, $0 < \theta_{\zeta 0} < 1$. Therefore, the upper bounds of e_ζ and $\tilde{W}_{\zeta f i}$ are calculated by $\|e_\zeta\| \leq \sqrt{2\vartheta_\zeta}$ and $\|\tilde{W}_{\zeta f i}\| \leq \sqrt{2a_\zeta \vartheta_\zeta}$. Then, $\|\tilde{d}_\zeta\| \leq \check{\sigma}_\zeta \|\tilde{W}_\zeta^T\| + \check{\varepsilon}_\zeta$ can be further obtained. According to fixed-time convergence characteristics of the RBFNN weight approximation errors $\tilde{W}_\zeta$, the observation errors $\tilde{d}_\zeta$ can converge into the upper bounds of $\|\tilde{d}_\zeta\|$ in fixed time. This completes the proof of Theorem 2.

6.3. Proof of Theorem 3

Consider Lyapunov functions $V_{\zeta 3}$ as follows:

$$V_{\zeta 3} = J_\zeta^*(\chi_\zeta) + \frac{1}{2} \tilde{\omega}_\zeta^T \tilde{\omega}_\zeta. \quad (34)$$

For simplification, let $\alpha_\zeta = q_\zeta \tilde{d}_\zeta$. It can be reasonably assumed that there exist positive constants k_ζ satisfying $\|\alpha_\zeta\| \leq k_\zeta$, according to Assumption 1, Claim in [40] and Eq. (3.1) in [35]. The time derivative of (34) yields:

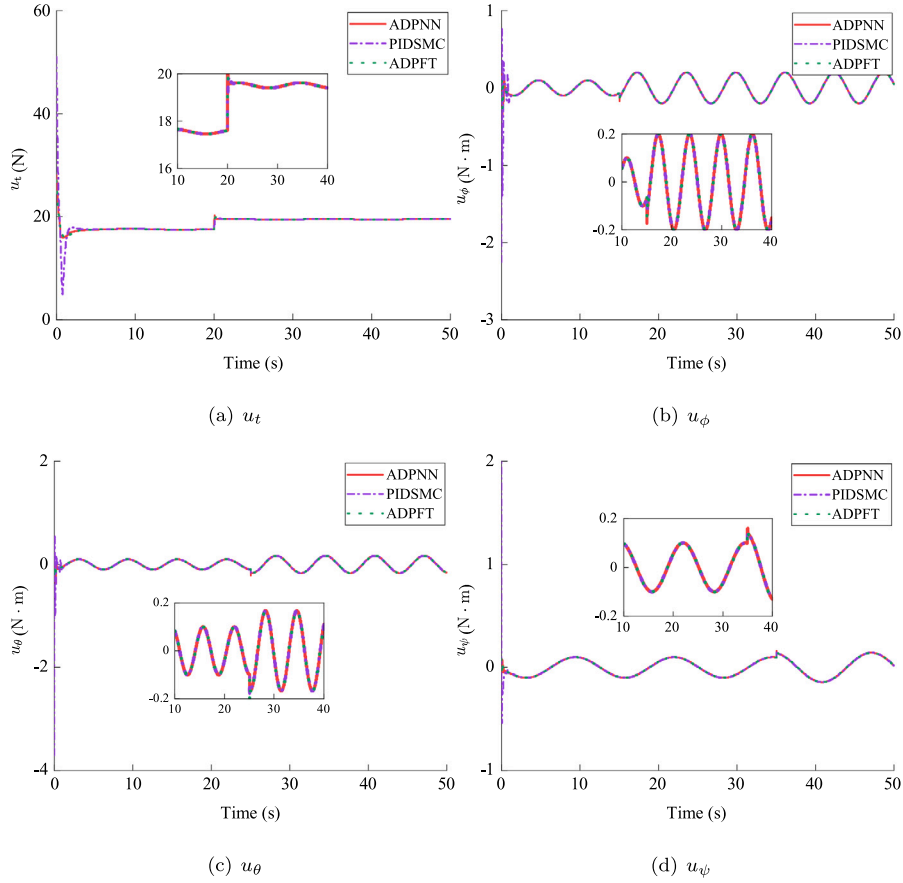


Fig. 7. Illustration of the four control inputs driven by different schemes.

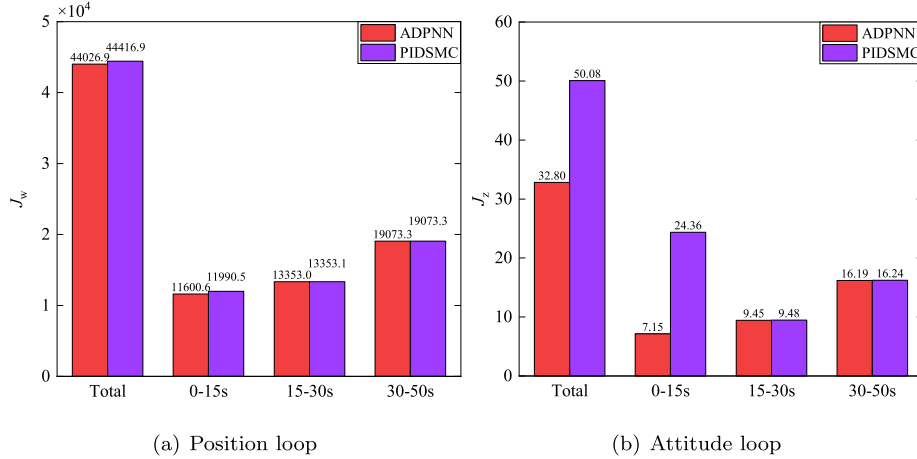


Fig. 8. The results of control cost driven by ADPNN and PIDSMC.

$$\begin{aligned}
\dot{V}_{\zeta 3} &\leq -\chi_{\zeta}^T Q_{\zeta} \chi_{\zeta} + \frac{1}{4} \nabla^T J_{\zeta}^* A_{\zeta} \nabla^T J_{\zeta}^* - \frac{1}{4} \nabla^T J_{\zeta}^* A_{\zeta} \nabla^T \delta_{\zeta} (\omega_{\zeta} + \tilde{\omega}_{\zeta}) \\
&\quad + \nabla^T J_{\zeta}^* \alpha_{\zeta} - \varpi_{\zeta} \nabla^T J_{\zeta} \tanh(\nabla J_{\zeta} / \rho_{\zeta}) + \varpi_{\zeta} \nabla^T J_{\zeta} \operatorname{erf}(\nabla J_{\zeta} / \rho_{\zeta}) \\
&\quad - \varpi_{\zeta} \nabla^T J_{\zeta}^* \operatorname{erf}(\nabla J_{\zeta} / \rho_{\zeta}) + \tilde{\omega}_{\zeta}^T \dot{\tilde{\omega}}_{\zeta} \\
&\leq -\chi_{\zeta}^T Q_{\zeta} \chi_{\zeta} - \tilde{\omega}_{\zeta}^T D_{\zeta} \tilde{\omega}_{\zeta} + P_{\zeta} \|\tilde{\omega}_{\zeta}\| + \epsilon_{\zeta z}, \tag{35}
\end{aligned}$$

where $\epsilon_{\zeta z} = \frac{\kappa_{\zeta 1} \epsilon_{\zeta H}^2}{2(1+\gamma_{\zeta}^T \gamma_{\zeta})} + \frac{\kappa_{\zeta 2} \tilde{\delta}_{\zeta}}{2} \|\omega_{\zeta}\|^2 + \frac{\epsilon_{\zeta}^2}{2} \lambda_{\max}(A_{\zeta}) + \varpi_{\zeta} \tilde{\epsilon}_{\zeta} (1 + \tilde{J}_{\zeta}) + Y_{\zeta}$ and $P_{\zeta} = \varpi_{\zeta} \tilde{\delta}_{\zeta} (1 + \tilde{J}_{\zeta})$ with $\tilde{J}_{\zeta} = \|\operatorname{erf}(\nabla J_{\zeta} / \rho_{\zeta})\|$. Moreover, Y_{ζ} are positive constants generated by the inequality related to the

$\tanh(\cdot)$ function. Additionally, the expressions D_{ζ} are $D_{\zeta} = \frac{\kappa_{\zeta 1} \gamma_{\zeta} \gamma_{\zeta}^T}{2(1+\gamma_{\zeta}^T \gamma_{\zeta})} + \frac{\kappa_{\zeta 2} \varphi_{\zeta}}{2} I_6 - \frac{G_{\zeta}}{4}$.

Assuming that there exist argument vectors $\Xi_{\zeta} = [\|\chi_{\zeta}\|, \|\tilde{\omega}_{\zeta}\|]^T$. The compact form of (35) can be expressed as the following inequalities

$$\dot{V}_{\zeta 3} \leq -\Xi_{\zeta}^T \check{D}_{\zeta} \Xi_{\zeta} + T_{\zeta} \Xi_{\zeta} + \epsilon_{\zeta z}, \tag{36}$$

with the terms $\check{D}_{\zeta}$, T_{ζ} expressed by $\check{D}_{\zeta} = \operatorname{diag}\{\lambda_{\min}(Q_{\zeta}), \lambda_{\min}(D_{\zeta})\}$ and $T_{\zeta} = [0, P_{\zeta}]$, respectively. It can be reasonably assumed that there are positive constants $\bar{T}_{\zeta}$ such that $\|T_{\zeta}\| \leq \bar{T}_{\zeta}$. Further, one has:

$$\dot{V}_{\zeta 3} \leq -\lambda_{\min}(\ddot{\mathbf{D}}_{\zeta}) \|\Xi_{\zeta}\|^2 + \bar{T}_{\zeta} \|\Xi_{\zeta}\| + \epsilon_{\zeta z}. \quad (37)$$

Then, $\dot{V}_{\zeta 3} < 0$ hold when the inequalities

$$\|\Xi_{\zeta}\| > \frac{\bar{T}_{\zeta} + \sqrt{\bar{T}_{\zeta}^2 + 4\lambda_{\min}(\ddot{\mathbf{D}}_{\zeta})\epsilon_{\zeta z}}}{2\lambda_{\min}(\ddot{\mathbf{D}}_{\zeta})}$$

are satisfied. This completes the proof.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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